

ITCS 3156 Final Project Report

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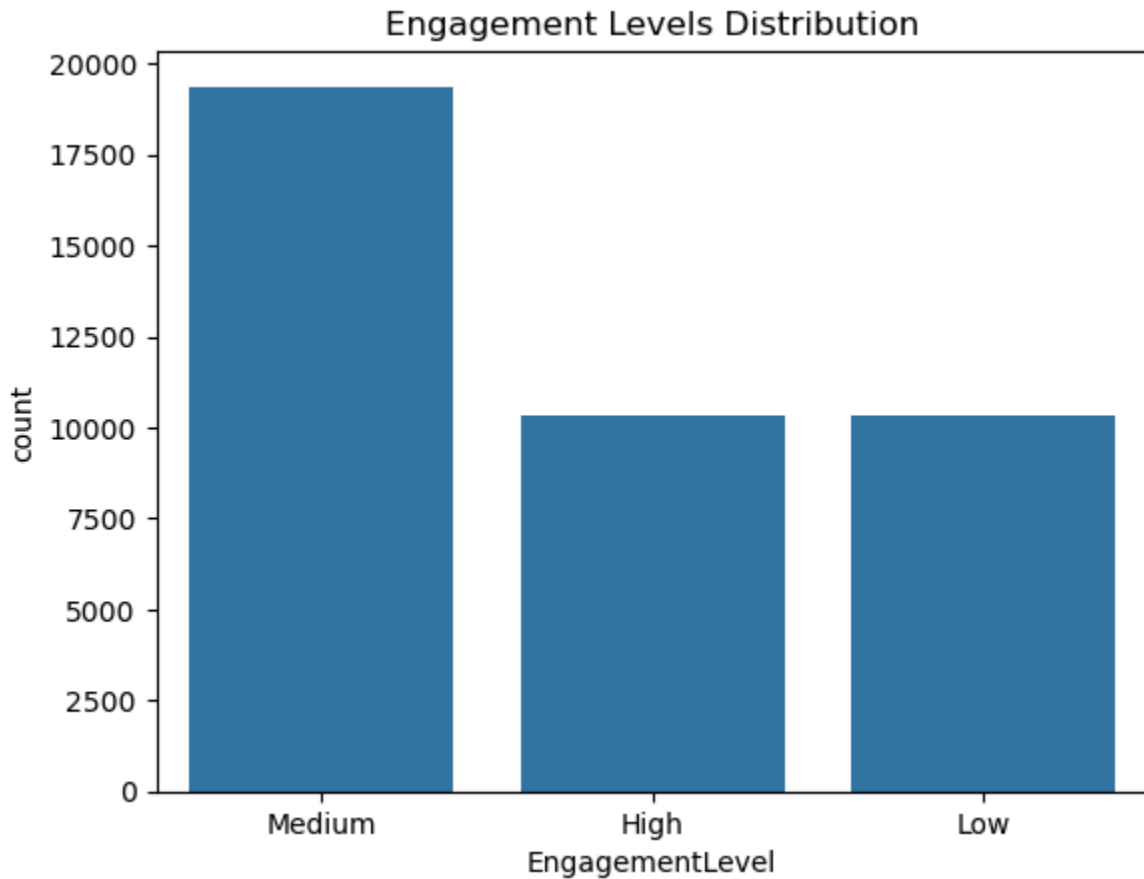
Introduction

The goal of this project is to train models to predict a certain player's engagement with a game based on their external factors. The dataset chosen is one found on Kaggle with variables such as Age, Gender, and Location of around 40,000 different people and information on the games they play. Using this, we should be able to have a model that can predict someone's retention on a certain game based on their information.

Data

Our target variable is "EngagementLevel", which is a variable split into three buckets – "Low", "Medium", and "High". This is a variable already included in the dataset, so this will be a supervised classification problem. We will have data to directly compare it to, so we will know if the model is incorrect or not. We have a mix of categorical and numeric variables; for categorical, we have Gender, Location, GameGenre, and GameDifficulty. For these variables (aside from gender, which was already encoded), we had to use one-hot encoding, as the model would not be able to distinguish the variables otherwise. For our numerical variables, we had PlayerID, Age, PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, and AchievementsUnlocked. We also had a boolean variable in InGamePurchases. Initially, we wanted to drop PlayerID, as well as the variables PlayTimeHours and SessionsPerWeek. PlayerID is irrelevant to our data, but PlayTimeHours and SessionsPerWeek would likely cause the model to simply tell us that a person will have high engagement if those respective numbers are high, which is not very useful. Before we got to that, we had to use local encoding to encode our target variable which had three different choices. Additionally, we mapped out the total outcomes of EngagementLevel, and found that the "medium" engagement accounted for the

majority of the cases.



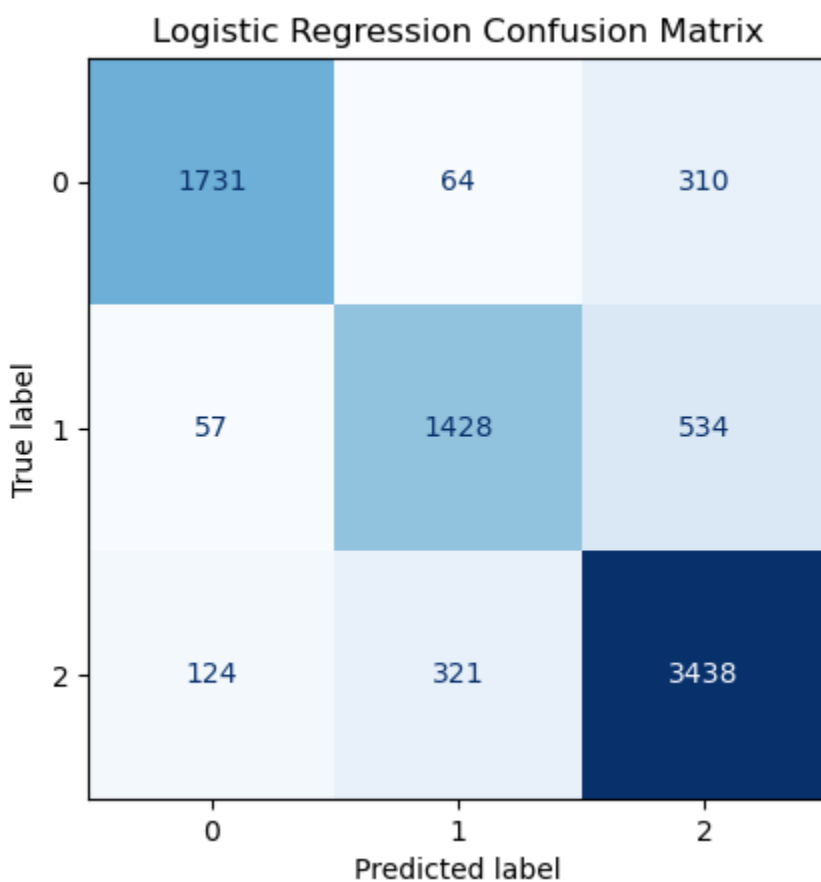
Methods

Due to the nature of this being a classification problem, we decided to go with a logistic regression model, as the tools such as feature importance and confusion matrices would be very useful in determining which predictors are most impactful on a player's engagement. It allows us to see a clear relationship between predictors and the target variable. Additionally, we also chose to use a K Nearest Neighbors model, as its simplicity would provide for easier explainability, as well as allowing us to detect similarities between players. KNN will be useful in determining the possible engagement level of someone based on other players who share similar factors. Due to the nature of KNN having multiple "nearest neighbors" to choose from, we created multiple models with different numbers for k. The point of this was to not only find the most accurate

model for this particular dataset, but to also make a wide range of models if we ever were to change the sample size.

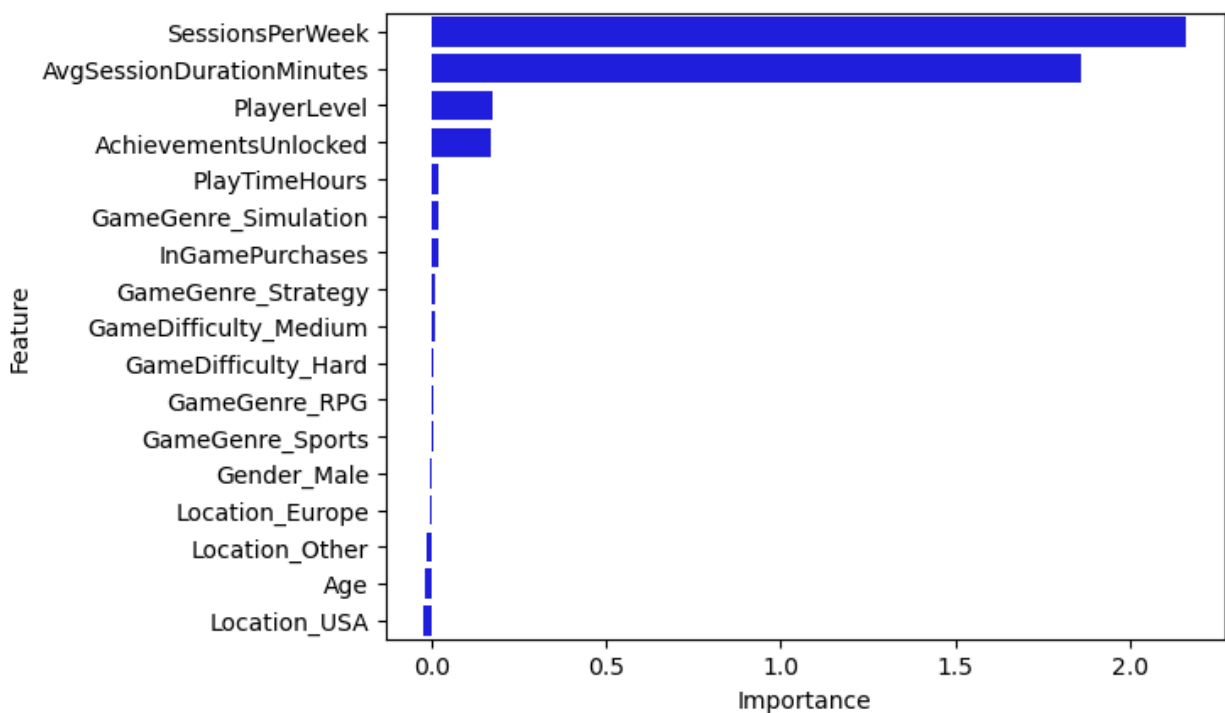
Results

Starting with the logistic regression model, when we included all features aside from PlayerID, the model had a respectable accuracy of 0.82. The majority of the mistakes were medium engagement levels being falsely predicted as high engagement levels, which interestingly did not happen on the lower side. This tells us that our model leans more towards predicting a high level of engagement over a smaller one.



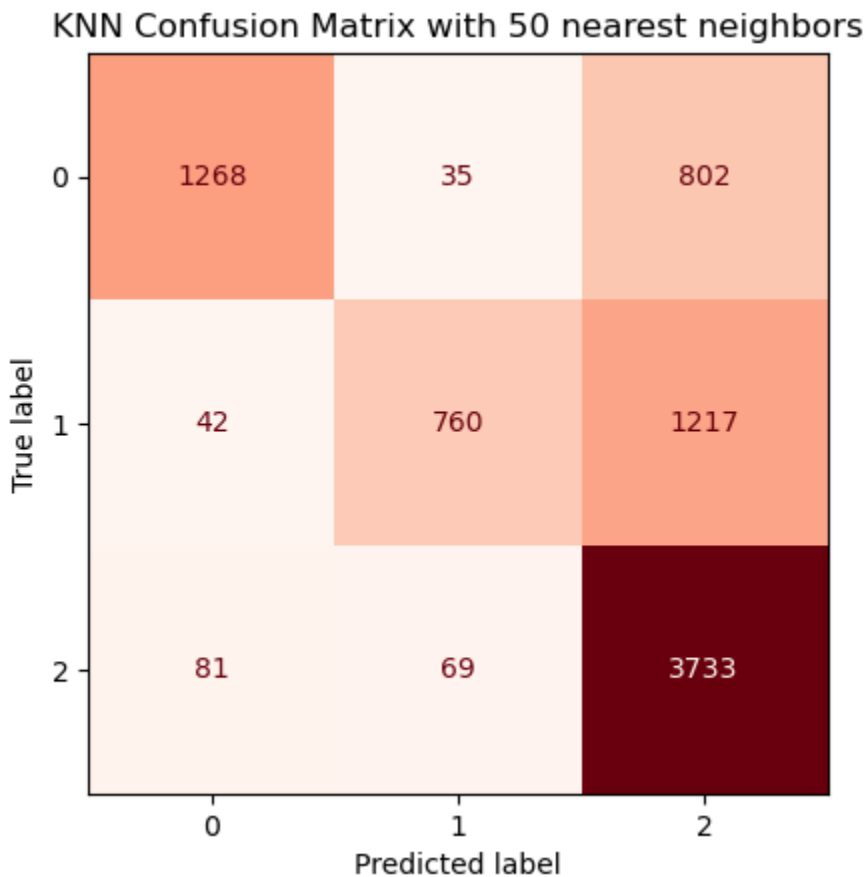
However, when looking at our feature importance, the model unsurprisingly reported back that the two most important features were SessionsPerWeek and AvgSessionDurationMinutes. This is

fine, but it left most of the other features in the dust, and doesn't really tell us what we wanted out of this dataset – what *kinds* of players are reaching high engagement levels – strategy game players? Higher difficulty level enjoyers? To fix this, we removed those top two features, and tried again. Unfortunately, we were met with disaster – the model had an accuracy of around 0.5, and the confusion matrix had a spread all over the place. While we were able to make a feature importance graph, the low accuracy level of the model makes that data mostly useless. We tried removing only one of those features, but then the model would become reliant on the other, and the same problem applies. This was a really unfortunate discovery, but it doesn't make this model useless. What we're finding is that engagement is mostly driven by how long and how often someone plays, and doesn't have much to do with what kind of game they are playing. In simpler words, the amount of engagement someone gets out of a game depends on how much they like it, and if they like it, they'll play it more often. At the end of the day, it's preference, and this model type wasn't able to garner much results out of player demographics.



Moving on to the K Nearest Neighbors model, the main goal was to create multiple of them to find the most accurate model, but also just have multiple ones in case the sample size changes. We created three models; one with 5 neighbors, one with 50 neighbors, and one with 75 neighbors. With 5 neighbors, the model outputted an accuracy of 0.68, with a huge lean towards predicting a high engagement rate. With 50 neighbors, the model outputted an accuracy of 0.72, with very high accuracy across the board; except when it came to a medium engagement level, where it only predicted about half correctly. With 75 neighbors, the model outputted an accuracy of 0.70, telling us that the sweet spot (for this particular training/test set) is somewhere between 50 and 75 neighbors. Using what we know from Logistic Regression, it's likely that our model heavily relies on playtime, so we can infer that players listed in the "low" category are a lot less closer to being in the "medium" category than the "high" engagement players are. In all three models, the left side of our confusion matrix (the model predicting a player in the "low" category) is extremely more scarce than the opposite side. This brings into question the dataset itself; did this dataset grab more hardcore gamers than casual gamers? Our preprocessing at the beginning showed that the low and high engagement buckets were pretty much equal, but we are unable to learn how close they were to being put in different categories. So, we can simply infer that the dataset's baseline is very high, and if you're really just a casual gamer, the model will

have no trouble predicting your engagement.



Conclusion

In conclusion, our models were unable to provide us insight on what kinds of infographics equate to high engagement levels, and instead equated high playtime levels with high engagement. Our logistic regression model struggled to find a relationship between engagement and game genres, ages, gender, etc. What this tells us is that game engagement doesn't necessarily have to do with external factors, such as where you live, nor does it have to do with the kind of game you're playing. The amount of engagement and enjoyment you will get out of a game simply comes down to preference – if you like a certain genre of games, then you'll most definitely accrue a lot of hours in the game, and therefore, engage with it more. Our two models may struggle to depict between an average game player and a hardcore gamer, but

can still determine if you just “casually” like a game or not, either based on features, or on a certain number of players similar to you.

Authors Note

While this project did not end with the result I was hoping for (likely due to the dataset I picked), I was still able to learn a lot about my coding skills with it. For starters, actually doing the process in a project makes it a lot less daunting. The coding portion of this project took me a lot less time than I expected – possibly due to me not having to do much feature engineering, as this dataset was already mostly project-ready. I think I could’ve learned more if I at least had to craft my own target variable, but I digress. Additionally, this project marks my first time uploading a real piece of code to GitHub, which is a huge step in my career. I may not be super proud of it, but I do have to start somewhere!

Dataset Link:

<https://www.kaggle.com/datasets/rabieelkharoua/predict-online-gaming-behavior-dataset/data>

GitHub Link: <https://github.com/afarmerd/ITCS31562>